

Direct comparisons require defining criteria for success, but in mechanistic interpretability, there is little agreement about the kinds of phenomena we should be measuring and precisely how they should be measured (with the exception of faithfulness, which is very common but still not standardized; Hanna, Pezzelle, and Belinkov 2024; Wang et al. 2023). Some tasks have started to integrate human/user evaluations, which will be especially useful for building interpretability tools that are grounded in real-world use cases and settings (Saphra et al. 2024). One benchmark that aims to enable direct comparison across mediator types and search methods is the Mechanistic Interpretability Benchmark (MIB; Mueller et al. 2025), which consists of two tracks: one for comparing circuits based on basis-aligned mediators, and another for comparing across mediator types/search methods. The fusion of these two tracks—i.e., comparing full causal graphs based on non-basis-aligned mediators, or even non-basis-aligned subspaces—could add value beyond the sum of these two separate tracks, and enable us to more directly benchmark progress on each of the goals of MI. Furthermore, Mueller et al. only measure (counterfactual) faithfulness and sparsity, as these are the most tractable metrics at the relatively large scale needed for a benchmark. Future benchmarks should develop more scalable measures of generality and selectivity (e.g., via out-of-distribution evaluations).

For model editing and localization, more task-specific downstream metrics and datasets will be needed. For example, Cohen et al. (2024) and Zhong et al. (2023) propose benchmarks to evaluate model editing methods on out-of-distribution examples, and Karvonen et al. (2024) propose to measure progress in feature disentanglement using board game models. Wu et al. (2025) define a measure for the quality of steering methods based on different mediator types and search methods. While not the main focus of this survey, we believe that building standardized benchmarks will be a key means to the end of assessing whether advancements in causal mediators are producing real improvements in applications of interpretability. More broadly, robust evaluation metrics and methods will lead to a more accurate science of the inner workings of language models, which will allow us to assess whether new causal abstractions are fundamentally more useful—for explaining the computations of a neural network, for verifying hypotheses, *and* for practical applications.

8. Conclusion

In any study analyzing model behaviors via analyzing model components, the type of component(s) analyzed will determine what kinds of findings are possible. Some units are more closely aligned to the target concepts, while others are more faithful to the model’s computation. Some units are easier to search over, but more difficult to understand (or vice versa). We have proposed a narrative and taxonomy of mechanistic interpretability research grounded in these units of analysis, or causal mediators. We have discussed the strengths and weaknesses of each mediator type, as well as what kinds of search methods are commonly employed for each. We have also discussed open problems in the field, focusing on those where this perspective reveals actionable and impactful research opportunities.

Appendix A: Types of interventions

To compute the indirect effect (Eq. 1), we must replace the value v_i of causal node V_i with some counterfactual value v'_i . Say we are using mediator type Z , and that we are performing an exhaustive search by computing the IE for all $z_i \in Z$, and then taking the

top components by IE. How should we compute z'_i ? There are many ways to derive z_i : some of these depend on x , others depend on whether x is a member of some class (e.g., inputs about dogs or inputs not about dogs), and others still depend on neither (e.g., are constant values or involve adding a noise term). Here, we briefly describe each of these classes of interventions, and describe how they will affect the kinds of components one will uncover.

Broadly, constant interventions will tell one which components have *any* impact on model behavior, regardless of how. Input-dependent interventions are more precise, but tend to have lower recall: they isolate components whose impact changes when the input changes in the specific way defined by the intervention. Class-dependent interventions are a sort of medium between these.

A.1 Input-dependent interventions

Deterministic interventions. If one cares about neurons that are sensitive to a specific contrast, then one can use input-dependent interventions (i.e., interventions where z'_i depends on x). For example, assume our target task is subject-verb agreement. Given an input $x = \text{"The key"}$, we want to locate neurons that increase the probability difference $m = p(\text{is}) - p(\text{are})$. We obtain z_i by running x through model $\mathcal{C}$ in a forward pass (denoted $\mathcal{C}(x)$) and storing the activation z_i of component(s) Z_i (which could be a neuron, for example). We then obtain z'_i by running $\mathcal{C}(x')$, where x' is a minimally different input that swaps the answer; here, x' would be *"The keys"*. This type of intervention preserves example-specific information, and varies only the grammatical number of the subject. This will only reveal neurons for which swapping grammatical number *and nothing else* will significantly affect the model's output.

Input-dependent interventions are precise: they reveal components targeted to a specific contrast between two prompts. However, humans must carefully curate controlled input pairs in which only one phenomenon is varied across x and x' . Input-dependent interventions work best for *binary* contrasts, where one defines two minimally different inputs that isolate neurons sensitive only to the difference between items in the pair. This yields counterfactuals that are semantically meaningful, making the results of an intervention easier to interpret. When working with categorical or ordinal variables, it is not immediately clear how to construct x' to recover all relevant components. Additionally, it does not recover all task-relevant components; it only recovers those sensitive to the contrast between x and x' . In other words, this is a low-recall method. For instance, Vig et al. (2020) enumerates through all possible gender pronouns and nouns related to a specific gender to measure gender bias; they note that full generalizability to all grammatical gender pronouns is difficult. Furthermore, such interventions can be privy to unreliable explanations. This was shown in Srivastava, Oikarinen, and Weng (2023), where the input data was corrupted to manipulate the concept assigned to a neuron. Hence, input-dependent interventions may require additional safeguards to ensure safety and fairness in critical real-life applications.

Stochastic interventions. Another common intervention type entails adding noise to z'_i , without defining some specific x'_i from which to derive it. For example, Meng et al. (2022, 2023) derive the counterfactual as $z'_i = z_i + \epsilon$, where $\epsilon \sim \mathcal{N}(0, 3\sigma_{Z_i})$. σ_{Z_i} is the standard deviation of Z_i on some dataset. This intervention depends on z_i by definition, but does not isolate a semantically meaningful contrast as input-dependent interventions do. This is conceptually closer to class- and independent-independent

interventions (§1.3), in that it will isolate components with *any* impact on the model’s behavior, regardless of the semantics of that impact. However, its stochasticity introduces variability in results and can be harder to interpret causally.

A.2 Class-dependent interventions

In contrast to input-dependent interventions, class-dependent interventions define a single intervention across a class of inputs. For example, Li, Davies, and Nadeau (2024) learn a mask over the computational graph of a language model to prevent the model from producing toxic content; here, the two classes are *toxic* and *not toxic*, and the intervention within one of those classes is the same as for all other inputs in that class.

Class-dependent interventions provide a single flexible intervention that works for any given input. However, they require a dataset of input-label pairs that can be used to learn the interventions. This requires labeled data, and is sensitive to spurious correlations. Furthermore, many labels we care about are hard to definitively label without ambiguity (e.g., bias or toxicity).

A.3 Class- and input-independent interventions

This type of intervention does not rely on the input nor a class label. The goal of these interventions is generally to fully remove (*ablate*) the information encoded by a mediator, regardless of whether the information is task-relevant.³⁰ A common ablation type is **zero ablations** (Dabkowski and Gal 2017; Lakretz et al. 2019; Geva et al. 2023), where the activation of a component is set to 0. This is not entirely principled, since 0 has no inherent meaning in an activation—for example, a neuron’s default activation may be non-zero, whereas 0 itself is out of distribution relative to what the model expects. A more principled ablation type is a **mean ablation** (Zeiler and Fergus 2014; Ghorbani and Zou 2020; McDougall et al. 2024), where the neuron’s activation is set to its mean value over some distribution—either task-specific data or general text data. A **resampling ablation** (Robnik-Šikonja and Kononenko 2008; Chan et al. 2022) is typically defined as a special case of mean ablations where the sample size is 1, and where the counterfactual input is randomly sampled.

Class- and input-independent interventions are a more general type of intervention that can be run without access to contrastive input/output pairs, and without labeled inputs. They allow us to tell whether *any* of the information in a mediator is necessary for a model to perform the task, but they may also affect other information in unanticipated ways; in other words, they have high recall and low precision relative to the previous intervention types. They may also cause performance on a task to drop in a way that reveals spurious mediators, rather than mediators that are conceptually relevant. For example, in subject–verb agreement, ablating a neuron that detects the word “dog” may reduce the probability of the correct verb form “is” over the incorrect verb form “are”, but this is a highly input-specific neuron that does not, in isolation, reveal general information about how models perform syntactic agreement.

³⁰ An ablation is a type of intervention. The goal of an ablation is to *remove* the information stored in a component. “Intervention” is a broader term that refers to setting some v_i to any value it would not naturally have taken.